

Formulation

$\exists \pi; \sum_{k=1}^4 M_k + P = H, \quad |H| = 14, \quad M_k \in \{\text{Chow}, \text{Pung}, \text{Kong}\}, \quad P = \text{Pair}$

H : a winning hand

M_k : the k -th meld (3 or 4 tiles)

- Chow: $(x, x+1, x+2)$ same suit
- Pung: (x, x, x)
- Kong: (x, x, x, x)

P : one pair (y, y)

A hand wins iff its 14 tiles can be partitioned into four valid melds plus one pair.
Special hands are uncovered, while it is easy to account for them.

Probabilistic winning of the player

Remark: Ignoring the winning of the opponents. We focus on finding the minimal rounds for the player to win, or a weighted combination of round numbers and rewards. Rewards are ignored for now.

The probability of winning at round t given current hand H_t and discard pool A_t is written as $\boxed{P_t(\text{WIN} \mid H_t)}$

$P(x_t \mid W_t); P(W_t \mid A_t, H_t); w_{a_t}(\pi \mid H_t, x_t); \mathbf{1}(\pi)$

where:

- $P(x_t \mid W_t)$: probability of drawing tile x_t from the remaining wall W_t
- $P(W_t \mid A_t, H_t)$: probability of the remaining wall given discard history and player hand
- $w_{a_t}(\pi \mid H_t, x_t)$: **deterministic optimization**, selecting the discard a_t that leads to the optimal next hand
- $\mathbf{1}(\pi)$: indicator of a valid winning formation

with $H_t^+ = (H_t \cup \{x_t\}) \setminus \{a_t\}$, $A_t = A_{t-1} \cup \{a_t\} \cup \{a_{t-1}, \text{opponents}\}$.

Remark: A_t is treated as a pool here. In principle, it is a time-ordered discard sequence conditioned on players.

One major difficulty is modeling $P(W_t \mid A_t, H_t)$.

Remark: Finding a winning formation π is a combinatorial optimization problem and may be impossible for certain (H_t, x_t) .

Probabilistic remaining wall

Consider $P(W_t \mid A_t, H_t)$.

The remaining wall is $W_t = T \setminus H_t \setminus A_t \setminus H_t^{\{i \neq \text{我}\}}$,

where:

- T : fixed total tile set
- H_t : known player hand
- A_t : known discard history
- $H_t^{\{i \neq \text{我}\}}$: unknown opponent hands

The problem reduces to the **opponent-hand likelihood** $P(H_t^{\{i \neq \text{我}\}} \mid H_t, A_t)$.

Remark: Modeling the opponent-hand likelihood is the core challenge.
It depends on assumptions about opponent playstyle.

Simplification 2: Opponents are independent and do not condition their decisions on each other.

Under this assumption, $P(H_t^{\{i \neq \text{我}\}} \mid H_t, A_t)$

$\prod_{i \neq \text{我}} P(H_t^i \mid A_t^i)$, where A_t^i is the discard history of opponent i .

Opponent hand inference (Bayesian update)

For a fixed opponent i , let $A_{1:t}^i = (a_1^i, \dots, a_t^i)$ denote the observed discard sequence.

The posterior over a hypothetical concealed hand H is updated sequentially via $\boxed{P(H \mid A_{1:t}^i) \propto P(a_t^i \mid H); P(H \mid A_{1:t-1}^i)}$

Remark: All modeling assumptions enter through the discard likelihood $P(a_t^i \mid H)$.

Toy discard likelihood model (average player)

We assume an average (non-expert) player whose discard choice is driven by hand efficiency and simple structure.

Approximate shanten (向听数)

Let $\widehat{h}(H)$ be a cheap approximation of shanten (向听数), measuring distance to a winning hand (4 melds + 1 pair).

Define the shanten change for discarding tile x : $\Delta h(x; H)$

$\widehat{h}(H \setminus \{x\}) - \widehat{h}(H)$. $\$ \$$

Only relative values of Δh are required.

Structural features

For tile $x \in H$:

- $v_{\text{struct}}(x; H)$: structural value
(0 = isolated 孤张, 1 = partial group 搭子, 2 = meld or pair 面子/对子)
 - $\mathbf{1}_{\text{main}}(x; H)$: indicator that x belongs to the main suit (主做花色)
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Discard score and likelihood

Define a discard energy: $E(x; H)$

$\alpha \Delta h(x; H) + \beta v_{\text{struct}}(x; H) + \gamma \mathbf{1}_{\text{main}}(x; H)$, $\alpha, \beta, \gamma > 0$. $\$ \$$

The discard likelihood is $\boxed{P(\text{discard } x \mid H)}$

$$\frac{\exp[-E(x; H)]}{\sum_{y \in H} \exp[-E(y; H)]} \quad \$ \$$$

This stochastic model favors discarding isolated, off-suit tiles while suppressing moves that increase shanten or destroy structure.

Optimal discard decision (one-step optimization)

At round t , after drawing a tile, the player holds a 14-tile hand H_t^+ and must choose a discard $a_t \in H_t^+$. $\$ \$$

We assume that the posterior distributions $P(H_t^i \mid A_t^i)$, $i \neq$ 我 over opponents' concealed hands are already available from the inference step.

Simplification 3: For now, we ignore opponent calling and deal-in risk, and consider only the effect of opponents through the remaining wall distribution.

Remaining wall distribution

Given the discard pool A_t , player hand H_t , and opponent posteriors, the expected remaining count of tile x in the wall is c_x

$$\mathbb{E}_{H_t^{i \neq \text{我}}} \left[\text{count of } x \text{ not in } H_t \cup A_t \cup H_t^{i \neq \text{我}} \right].$$

The draw probability is approximated by $P(x \mid W_t)$

$$\frac{c_x}{\sum_y c_y}.$$

Hand value function

Remark: We define a cheap hand value function $V(H)$ that approximates the probability of winning soon. It favors ready hands, hands with many available waits, and hands closer to completion.

Let:

- $\widehat{h}(H)$ be the approximate shanten (向听数),
- $\mathcal{W}(H)$ be the wait set (等牌集合), i.e. tiles that complete the hand.

Define $V(H)$

$$\mathbf{1}(\widehat{h}(H)=0); \sum_{x \in \mathcal{W}(H)} P(x \mid W_t); +; \mathbf{1}(\widehat{h}(H)>0); \exp(-\lambda \widehat{h}(H)), \quad \lambda > 0.$$

One-step expectimax evaluation

For each discard candidate a_t , define $H_t^- = H_t + \setminus a_t$.

The expected value of discarding a_t is approximated by $Q(a_t)$

$$\sum_x P(x \mid W_t); V(H_t^- \cup \{x\})$$

This corresponds to a one-step lookahead over the next draw, marginalizing over wall uncertainty.

The optimal discard decision is $\boxed{a_t^*}$

$$\arg \max_{a_t \in H_t^+} Q(a_t)$$

This strategy maximizes an effective winning probability while implicitly favoring earlier wins.

Remarks: Summarizing the above optimizing strategy: 1. this is a myopic approximation to the optimal POMDP policy; 2. Opponent posteriors affect decisions only through $P(x \mid W_t)$ at this stage; 3. The framework is fully extensible: deal-in risk (放炮风险) and opponent winning probability can be added as penalty terms to $Q(a_t)$ in later refinements.

